PSRI HOSPITAL
A MULTI SPECIALITY HOSPITAL

Avoiding Total Knee Replacement: How Robotic Microplasty Transforms

Microplasty Knee Surgery Made Precise with Robotic Technology can avoid need for total knee replacement in most of patients.

Not all knee arthritis affects the entire knee. In many patients, damage is limited to only one part. For such cases, a partial knee replacement—where only the worn-out portion is changed—can be an excellent solution.



Partial knee replacement is a smaller surgery, but it actually demands precision with greater skill and accuracy. The healthy parts of the knee including all ligaments must be preserved, and the new implant must fit perfectly to maintain natural movement. This level of precision is difficult to achieve consistently without advanced technology. This was a major limitation for using this procedure widely.

This is where Mako Robotic assistance makes a major difference. Before surgery, a 3-D CT scan of the patient's knee is used to create a detailed surgical plan. The robotic system uses the patient's own knee landmarks and anatomy to guide the surgeon during the operation, helping recreate the knee's natural movement.

Because of this accuracy, the surgery requires a smaller incision, causes less tissue damage and blood loss, and allows for faster recovery. Most of the natural knee is saved, leading to better function and more natural walking.

The goal of robotic partial knee replacement is to create a "forgotten joint"—a knee that feels so natural that patients forget they ever had a joint replacement and can return to everyday life with confidence.

- High Accuracy
- Quick Recovery
- Less Pain
- Minimal Blood Loss

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